

SARH, Chad

WMO: 647500

Lat: 9.150N

Lon: 18.383E

Elev: 1198

StdP: 14.07

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	60.0	62.4	24.0	19.0	90.4	27.1	22.0	89.9	17.1	78.7	14.4	78.4	1.5	320	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	21.5	105.5	69.5	104.0	69.7	102.5	69.9	79.4	89.2	78.7	88.6	78.2	88.1	6.6	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.3	148.4	82.8	76.6	145.2	82.4	75.6	140.1	82.0	44.1	89.1	43.3	89.1	42.6	88.0	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.4	13.9	12.0	DB	55.5	108.3	3.3	0.9	53.1	108.9	51.2	109.4	49.3	109.9	46.8	110.6	
			WB	46.2	83.6	1.7	1.8	44.9	84.9	43.9	86.0	42.9	87.0	41.6	88.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	83.5	80.0	85.7	89.9	90.6	87.3	83.1	80.4	79.6	80.4	82.3	82.7	80.1	(6)
(7)		DBStd	4.91	3.97	3.78	2.84	3.43	4.15	3.39	2.99	2.42	2.44	2.33	2.41	2.86	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	12222	930	998	1236	1217	1158	993	943	917	913	1003	980	934	(10)
(11)		CDD65	6747	465	578	771	767	693	543	478	452	463	538	530	469	(11)
(12)		CDH74	81121	5863	8027	11576	11458	9385	6075	4297	3548	3948	5317	6026	5601	(12)
(13)		CDH80	45553	3540	5105	7599	7335	5389	2819	1643	1200	1539	2505	3488	3391	(13)
(14)	Wind	WSAvg	4.4	4.5	4.5	5.1	6.4	6.2	5.1	4.1	3.9	3.7	3.4	3.0	3.5	(14)
(15)	Precipitation	PrecAvg	42.9	0.0	0.0	0.3	1.3	4.8	6.0	9.5	9.7	7.7	2.9	0.1	0.0	(15)
(16)		PrecMax	68.6	0.0	0.7	2.0	4.5	18.1	12.0	25.5	25.2	20.4	7.8	1.2	0.2	(16)
(17)		PrecMin	30.1	0.0	0.0	0.0	0.0	0.9	1.2	2.8	3.3	3.4	0.0	0.0	0.0	(17)
(18)		PrecStd	8.7	0.0	0.2	0.5	1.3	3.5	2.4	4.4	4.1	3.7	1.9	0.3	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	102.5	106.2	107.4	106.9	104.3	98.9	94.8	92.7	93.5	97.1	100.2	100.6	(19)
MCWB			65.7	67.3	69.2	71.2	74.6	76.6	76.7	78.0	77.8	74.3	70.3	66.1	(20)	
2%		DB	100.6	104.3	105.6	104.6	102.2	96.3	92.1	89.9	91.6	95.1	98.4	98.7	(21)	
		MCWB	65.2	66.5	68.6	73.0	75.0	76.6	77.0	77.1	77.9	75.2	69.8	66.6	(22)	
5%		DB	98.7	102.6	104.2	103.1	100.2	93.9	90.0	88.2	89.9	93.3	96.8	97.1	(23)	
		MCWB	64.1	66.1	68.7	73.6	75.1	76.4	76.7	76.6	77.4	76.3	70.2	65.9	(24)	
10%		DB	95.9	100.7	102.6	101.3	97.5	91.7	88.1	86.3	88.1	91.4	95.2	95.3	(25)	
		MCWB	63.0	65.6	68.4	73.8	75.1	76.3	76.2	76.4	76.8	76.5	70.8	65.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.7	74.3	77.6	78.5	79.9	79.9	79.2	79.4	80.1	80.1	78.6	75.8	(27)
MCD B			93.3	97.8	96.6	95.6	93.3	91.2	88.0	86.7	88.0	89.4	91.3	92.5	(28)	
2%		WB	67.9	72.0	75.7	77.3	78.2	78.6	78.2	78.3	79.0	79.1	77.1	72.6	(29)	
		MCD B	93.5	94.8	96.5	95.2	91.8	89.3	87.1	85.9	87.5	88.5	89.8	87.0	(30)	
5%		WB	65.8	70.5	74.5	76.4	77.3	78.0	77.6	77.7	78.3	78.3	75.9	69.8	(31)	
		MCD B	93.0	93.5	95.5	94.6	91.3	88.7	86.5	85.2	86.6	87.7	88.4	87.6	(32)	
10%		WB	64.1	68.5	73.3	75.6	76.6	77.3	77.0	77.0	77.6	77.6	74.7	67.5	(33)	
		MCD B	91.3	93.2	94.2	93.9	90.7	88.0	85.8	84.4	85.7	87.0	86.8	87.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	33.2	31.2	26.7	21.5	19.1	16.9	14.8	14.4	16.2	18.9	27.3	32.5	(35)
MCDBR			34.6	32.4	28.9	23.6	21.9	19.7	17.9	16.9	18.6	21.8	29.2	33.2	(36)	
MCWBR			11.7	11.7	9.9	6.9	6.2	6.5	6.6	6.9	7.6	7.0	7.8	10.1	(37)	
5% WB		MCDBR	34.0	28.7	23.4	21.2	19.7	17.8	16.3	16.0	17.6	19.2	23.9	31.3	(38)	
		MCWBR	12.5	12.1	8.5	6.9	6.6	7.2	6.9	7.2	7.9	7.7	7.7	11.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.450	0.549	0.626	0.594	0.547	0.522	0.489	0.478	0.487	0.507	0.482	0.455	(40)	
taud		2.134	1.855	1.724	1.843	1.984	2.077	2.203	2.245	2.203	2.122	2.119	2.122	(41)		
Ebn at Noon		272	249	232	237	243	246	256	263	263	257	261	267	(42)		
Edh at Noon		49	67	77	68	58	52	46	45	47	51	49	49	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1931	1985	2012	1975	1865	1705	1569	1523	1672	1819	1902	1862	(44)	
(45)		RadStd	62	74	107	93	66	71	71	84	70	75	61	31	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page